Year 1 AS Statistics and Mechanics

Chapter 10

Forces and motion

Practice Booklet

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10.1 Force diagrams

Practice Book, printed page 67.

1. A crate is suspended motionless from a rope. Draw a force diagram to show all the forces acting on the crate.

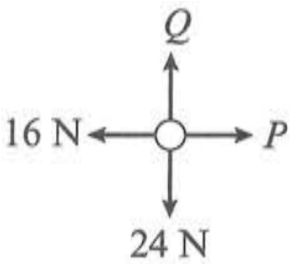
2. A particle of weight 12 N sits at rest on a horizontal plane. State the value of the normal reaction acting on the particle.

3. Given that each of the particles is stationary, work out the values of P and Q:

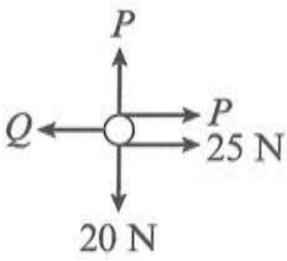
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c

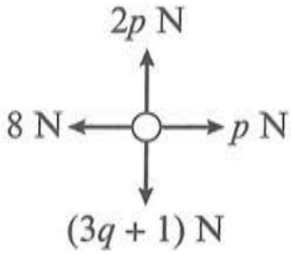


Hint

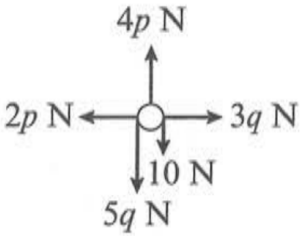
When a particle is stationary, the forces are all balanced.

4. P Each diagram shows a particle at rest acted on by a set of forces. The particle is at rest in each case. In each case, find the values of p and q.

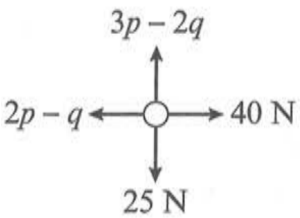
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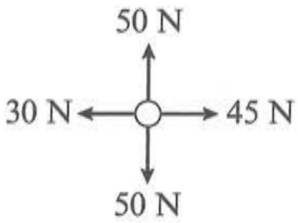


5. Each diagram shows the forces acting on a particle.

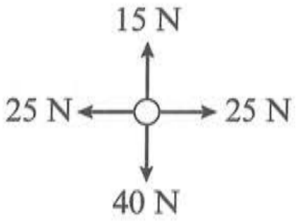
i. Work out the size and direction of the resultant force.

ii. Describe the motion of the particle.

a



b



6. A car with weight 20000 N is moving along a horizontal level road. The car's engine provides a forward thrust of 6000 N. The total resistance is modelled as a constant force of magnitude 800 N.

a. By modelling the car as a particle, draw a force diagram to show the forces acting on the car.

b. Calculate the resultant force acting on the car.

7. P A lorry is moving along a horizontal level road. The lorry's engine provides a constant driving force. The motion of the lorry is opposed by a constant resistance.

a. Modelling the lorry as a particle, draw a force diagram to show the forces acting on the lorry.

b. Given that the resultant force acting on the lorry is 7500 N in the direction of motion, and that the magnitude of the driving force is three times the magnitude of the resistance force, calculate the magnitude of the resistance.

10.2 Forces as vectors

Practice Book, printed page 68.

1. In each part of the question a particle is acted upon by the forces given. Work out the resultant force acting on the particle.

a. $(2\mathbf{i} - 4\mathbf{j})$ N and $(-\mathbf{i} + 5\mathbf{j})$ N

b. $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ N and $\begin{pmatrix} -2 \\ -5 \end{pmatrix}$ N

c. $(\mathbf{i} - 2\mathbf{j})$ N, $(4\mathbf{i} - 2\mathbf{j})$ N and $(-2\mathbf{i} + 6\mathbf{j})$ N

d. $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$ N, $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ N and $\begin{pmatrix} 3 \\ -8 \end{pmatrix}$ N

Hint

You can find the resultant force by adding the vectors.

2. The forces $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ N, $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ N and $\begin{pmatrix} a \\ b \end{pmatrix}$ N act on an object which is in equilibrium. Find the values of a and b.

Hint

If an object is in equilibrium, then the resultant force is zero.

3. The forces $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ N, $\begin{pmatrix} 5 \\ -6 \end{pmatrix}$ N, $\begin{pmatrix} -6 \\ 4 \end{pmatrix}$ N and $\begin{pmatrix} a \\ b \end{pmatrix}$ N act on an object which is in equilibrium. Find the values of a and b.

4. The forces $(a\mathbf{i} + b\mathbf{j})$ N, $(8\mathbf{i} + 2b\mathbf{j})$ N and $(-3a\mathbf{i} - 8\mathbf{j})$ N act on an object which is in equilibrium. Find the values of a and b.

5. For each of the following forces find:

i. the magnitude of the force

ii. the angle the force makes with the unit vector $\mathbf{i}$.

a. $(5\mathbf{i} + 2\mathbf{j})$ N

b. $(4\mathbf{i} - 3\mathbf{j})$ N

c. $\begin{pmatrix} -5 \\ 4 \end{pmatrix}$ N

d. $\begin{pmatrix} -6 \\ -2 \end{pmatrix}$ N

Hint

Draw a triangle of forces. Use Pythagoras' theorem to find the magnitude of the force and use trigonometry to find the angle.

6. In this question, $\mathbf{i}$ represents the unit vector due east, and $\mathbf{j}$ represents the unit vector due north. A particle is acted upon by forces of:

a. $(5\mathbf{i} - 3\mathbf{j})$ N, $(2\mathbf{i} + \mathbf{j})$ N and $(-3\mathbf{i} - 5\mathbf{j})$ N

b. $(3\mathbf{i} + 2\mathbf{j})$ N, $(-4\mathbf{i} - 5\mathbf{j})$ N and $(4\mathbf{i} + 2\mathbf{j})$ N

In each case, work out:

i. the resultant vector

ii. the magnitude of the resultant vector

iii. the bearing of the resultant vector.

7. E/P The forces $\mathbf{F}_1 = \begin{pmatrix} 2a \\ b \end{pmatrix}$ N, $\mathbf{F}_2 = \begin{pmatrix} b \\ -a \end{pmatrix}$ N and $\mathbf{F}_3 = \begin{pmatrix} -11 \\ 1 \end{pmatrix}$ N act on an object which is in equilibrium.

a. Find the values of a and b. (3 marks)

The force $\mathbf{F}_3$ is removed. The resultant of the forces $\mathbf{F}_1$ and $\mathbf{F}_2$ is $\mathbf{R}$. (2 marks)

b. Calculate, in newtons, the magnitude of $\mathbf{R}$.

c. Calculate, to the nearest degree, the angle between the line of action of $\mathbf{R}$ and the unit vector $\mathbf{i}$. (3 marks)

8. E/P The forces $\mathbf{F}_1 = (4a\mathbf{i} + a\mathbf{j})$ N, $\mathbf{F}_2 = (-2b\mathbf{i} - 8b\mathbf{j})$ N and $\mathbf{F}_3 = (-12\mathbf{i} + 12\mathbf{j})$ N act on an object which is in equilibrium.

a. Find the values of a and b. (3 marks)

The force $\mathbf{F}_3$ is removed. The resultant of the forces $\mathbf{F}_1$ and $\mathbf{F}_2$ is $\mathbf{R}$.

b. Calculate, in newtons, the magnitude of $\mathbf{R}$. (2 marks)

c. Calculate, to the nearest degree, the angle between the line of action of $\mathbf{R}$ and the unit vector $\mathbf{j}$. (3 marks)

9. E/P A particle is acted upon by two forces $\mathbf{F}_1$ and $\mathbf{F}_2$, given by $\mathbf{F}_1 = (-4\mathbf{i} + 7\mathbf{j})$ N and $\mathbf{F}_2 = (2a\mathbf{i} - a\mathbf{j})$ N, where a is a positive constant. The resultant of $\mathbf{F}_1$ and $\mathbf{F}_2$ is $\mathbf{R}$.
Given that $\mathbf{R}$ is parallel to $3\mathbf{i} + \mathbf{j}$, find the value of a. (4 marks)

10.3 Forces and acceleration

Practice Book, printed page 69.

1. Find the acceleration when a particle of mass 250 kg is acted on by a resultant force of 160 N.

Hint

Use Newton's second law of motion: $F = ma$

2. A 6 kg object accelerates at a rate of 4 m s^{-2} when a force is applied. Find the magnitude of the force.

3. Find the weight, in newtons, of a particle of mass 9 kg.

Hint

$W = mg$, where g is the acceleration due to gravity.

4. A box with mass 3 kg is at rest on a smooth table.
The box is pulled across the table by a horizontal string.
The tension in the string has magnitude 0.9 N. Find:

a. the acceleration of the box

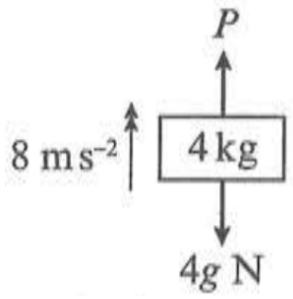
b. the time taken to pull the box 15 m.

Hint

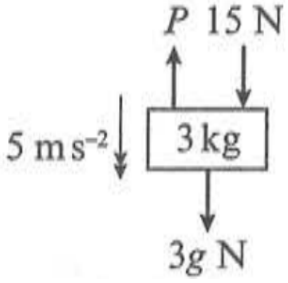
In part **b**, use one of the *suvat* formulae. ← Section 9.4

5. In each scenario, the forces acting on the body cause it to accelerate as shown.
In each case, find the magnitude of the unknown force, P.

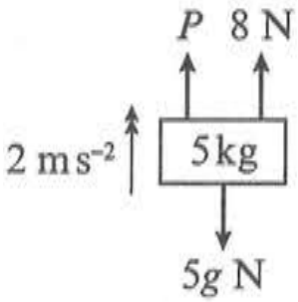
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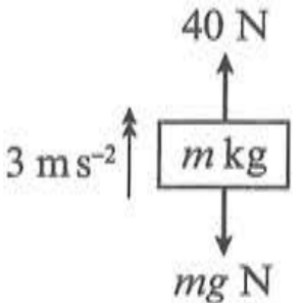


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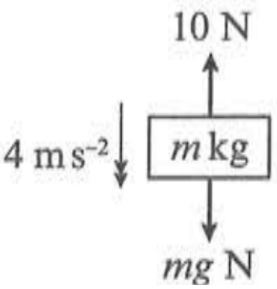


6. In each situation, the forces acting on the body cause it to accelerate as shown.
Find the mass, m kg, of the body in each case.

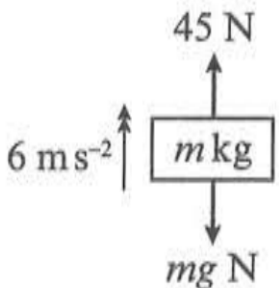
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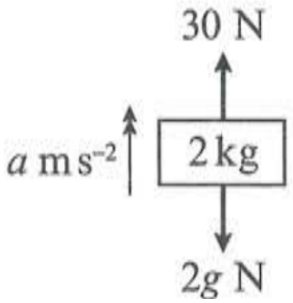


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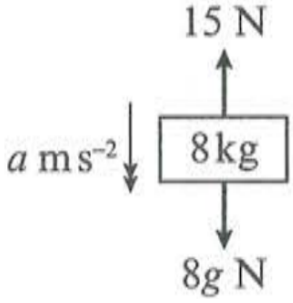


7. In each situation, the forces acting on the body cause it to accelerate as shown with magnitude a m s^{-2}. In each case find the value of a.

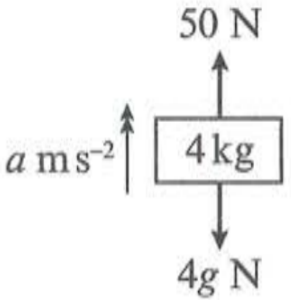
a



b



c



8. P A horizontal force of 40 N acts upon a particle of mass 5 kg, causing it to accelerate at 4 m s^{-2} along a rough horizontal plane. A constant resistive force opposes the motion of the particle. Calculate the magnitude of the resistive force.

9. An air-sea rescue crew member is suspended from a helicopter by a cable. The crew member has mass 60 kg. Find the tension in the cable when the crew member is being raised:

a. at a constant speed

b. with an acceleration of 0.5 m s^{-2}.

10. E/P A lorry with mass 6000 kg accelerates uniformly from rest to a speed of 12 m s^{-1} in 10 seconds along a straight horizontal road. The lorry experiences a constant resistive force of 1800 N. Find:

a. the acceleration of the lorry (3 marks)

b. the driving force provided by the engine of the lorry. (2 marks)

11. E/P A trolley with mass 60 kg is pulled in a straight line along a horizontal path by means of a horizontal rope attached to its front end. The trolley moves with a constant acceleration of 3.5 m s^{-2}. The tension in the rope is 330 N and the trolley is subject to a constant resistive force of magnitude k N.

a. Find the value of k. (3 marks)

At the instant the trolley is moving with speed 3 m s^{-1}, the rope breaks.

b. Calculate how much further the trolley moves before coming to rest. (3 marks)

10.4 Motion in 2 dimensions

Practice Book, printed page 70. In all the questions in this exercise, **i** represents the unit vector due east and **j** represents the unit vector due north.

1. A resultant force of $(6\mathbf{i} + \mathbf{j})$ N acts upon a particle of mass 4 kg.

a. Find the acceleration of the particle in the form $(p\mathbf{i} + q\mathbf{j})$ m s^{-2}.

b. Find the magnitude and bearing of the acceleration of the particle.

Hint

Use $\mathbf{F} = m\mathbf{a}$

2. A resultant force of $(7\mathbf{i} + 2\mathbf{j})$ N acts on a particle of mass m kg causing it to accelerate at $(28\mathbf{i} + 8\mathbf{j})$ m s^{-2}. Work out the mass of the particle.

3. A particle of mass 5 kg is acted on by a force $\mathbf{F}$.
Given that the particle accelerates at $(10\mathbf{i} - 6\mathbf{j})$ m s^{-2}, find:

a. an expression for $\mathbf{F}$ in the form $(p\mathbf{i} + q\mathbf{j})$ N

b. the magnitude and bearing of $\mathbf{F}$.

Hint

Mass is a scalar quantity. This means that the acceleration of an object will be in the same direction as the line of action of the resultant force acting on the object.

4. Two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$, act on a particle of mass m.
Find the acceleration of the particle, given that:

a. $\mathbf{F}_1 = (3\mathbf{i} + 8\mathbf{j})$ N, $\mathbf{F}_2 = (-2\mathbf{i} + 3\mathbf{j})$ N, $m = 0.5$ kg

b. $\mathbf{F}_1 = (8\mathbf{i} - 6\mathbf{j})$ N, $\mathbf{F}_2 = (3\mathbf{i} + 4\mathbf{j})$ N, $m = 7$ kg

c. $\mathbf{F}_1 = (-50\mathbf{i} - 40\mathbf{j})$ N, $\mathbf{F}_2 = (35\mathbf{i} + 15\mathbf{j})$ N, $m = 12$ kg

d. $\mathbf{F}_1 = 8\mathbf{i}$ N, $\mathbf{F}_2 = (-5\mathbf{i} - 10\mathbf{j})$ N, $m = 2.5$ kg

5. A particle of mass 5 kg is initially at rest. It is acted on by three forces: $\mathbf{F}_1 = (2\mathbf{i} + 5\mathbf{j})$ N, $\mathbf{F}_2 = (-\mathbf{i} - 6\mathbf{j})$ N and $\mathbf{F}_3 = (3\mathbf{i} - 2\mathbf{j})$ N. Find:

a. the magnitude and direction of the acceleration of the particle

b. the time taken for the particle to reach a speed of 2 m s^{-1}

c. the speed of the particle after it has travelled 10 m.

Hint

For parts **b** and **c**, find the magnitude of the acceleration vector, then use the *suvat* formulae.

6. E/P Two forces, $\begin{pmatrix} 1 \\ -5 \end{pmatrix}$ N and $\begin{pmatrix} p \\ q \end{pmatrix}$ N, act on a particle P.
The resultant of the two forces is $\mathbf{R}$. Given that $\mathbf{R}$ acts in a direction which is parallel to the vector $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, show that $2p + q - 3 = 0$. (3 marks)

7. E/P A particle of mass 2.5 kg is being acted upon by three forces, $\mathbf{F}_1 = (3\mathbf{i} + 8\mathbf{j})$ N, $\mathbf{F}_2 = (-7\mathbf{i} - 10\mathbf{j})$ N and $\mathbf{F}_3 = (p\mathbf{i} + q\mathbf{j})$ N. Given that the particle is accelerating at a rate of $(1.6\mathbf{i} - 2.4\mathbf{j})$ m s^{-2}, find the values of p and q. (3 marks)

8. E A particle of mass 3 kg starts from rest and is acted upon by a force $\mathbf{R}$ of $(9\mathbf{i} + b\mathbf{j})$ N.
$\mathbf{R}$ acts on a bearing of 060°.

a. Find the value of b. (1 mark)

b. Calculate the magnitude of $\mathbf{R}$. (2 marks)

c. Work out the magnitude of the acceleration of the particle. (2 marks)

d. Find the time taken for the particle to travel a distance of 75 m. (3 marks)

9. E/P Three forces act on a particle: $\mathbf{F}_1 = (-4\mathbf{i} + 5\mathbf{j})$ N, $\mathbf{F}_2 = (-\mathbf{i} + 7\mathbf{j})$ N and $\mathbf{F}_3 = (p\mathbf{i} + q\mathbf{j})$ N.

a. Given that this particle is in equilibrium, determine the values of p and q. (2 marks)

Force $\mathbf{F}_2$ is removed.

b. Given that in the first 10 seconds of its motion the particle travels a distance of 130 m, find the exact mass of the particle, in kg. (3 marks)

10. E/P A particle of mass m kg is acted upon by forces of $(10\mathbf{i} + 8\mathbf{j})$ N, $(4\mathbf{i} + 12\mathbf{j})$ N and $(-2\mathbf{i} - 4\mathbf{j})$ N, causing it to accelerate at 5 m s^{-2}. Work out the mass of the particle. (3 marks)

11. E/P Two forces, $(6\mathbf{i} - 2\mathbf{j})$ N and $(p\mathbf{i} + q\mathbf{j})$ N, act on a particle of mass m kg. The resultant of the two forces is $\mathbf{R}$. Given that $\mathbf{R}$ acts in a direction which is parallel to the vector $(2\mathbf{i} + \mathbf{j})$,

a. find the angle between $\mathbf{R}$ and the vector $\mathbf{j}$ (2 marks)

b. show that $p - 2q + 10 = 0$. (4 marks)

c. Given that $p = 4$ and that the particle moves with an acceleration of magnitude $2\sqrt{5}$ m s^{-2}, find the value of m. (7 marks)

10.5 Connected particles

Practice Book, printed page 72.

1. Two particles P and Q, of masses 6 kg and 10 kg respectively, are connected by a light inextensible string. The particles are on a smooth horizontal plane. A horizontal force of magnitude F is applied to P in a direction away from Q and when the string is taut the particles move with acceleration 0.4 m s^{-2}.

a. Find the value of F.

b. Find the tension in the string.

Hint

In part **a**, by considering the system as a single particle you eliminate the need to find the tension in the string.
In part **b**, the particles need to be considered separately to find the tension in the string.

2. Two particles P and Q, of masses 8 kg and m kg respectively, are connected by a light inextensible string. The particles are on a smooth horizontal plane. A horizontal force of 50 N is applied to Q in a direction away from P. When the string is taut the particles move with acceleration 2.5 m s^{-2}.

a. Find the value of m.

b. Find the tension in the string.

3. Two particles P and Q, of masses 15 kg and 25 kg respectively, are connected by a light inextensible rod. The particles lie on a smooth horizontal plane. A horizontal force of 80 N is applied to Q in a direction towards P, causing the particles to move with acceleration a m s^{-2}.

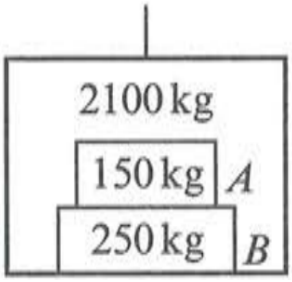
a. Find the value of a.

b. Find the thrust in the rod.

Hint

The fact that the rod is inextensible means that both particles have the same acceleration.

4. Two boxes A and B, of masses 150 kg and 250 kg respectively, sit on the floor of a lift of mass 2100 kg. Box A rests on top of box B. The lift is supported by a light inextensible cable and is descending with constant acceleration 1.2 m s^{-2}.



a. Find the tension in the cable.

b. Find the force exerted by box B on:

i. box A

ii. the floor of the lift.

5. A lorry of mass $2m$ kg is towing a trailer of mass $5m$ kg along a straight horizontal road. The lorry and trailer are connected by a light inextensible tow-bar. The lorry exerts a driving force of 60000 N causing the lorry and trailer to accelerate at 4 m s^{-2}. The lorry and trailer experience resistances of 3200 N and 12000 N respectively.

a. Find the mass of the lorry and hence the mass of the trailer.

b. Find the tension in the tow-bar.

6. E/P Two particles A and B, of masses 16 kg and 9 kg respectively, are connected by a light inextensible string. Particle B hangs directly below particle A. A force of 250 N is applied vertically upwards to particle A, causing the particles to accelerate.

a. Find the value of P. (3 marks)

b. Find the tension in the string. (2 marks)

7. E/P Two particles A and B, of masses 15 kg and 12 kg respectively, are connected by a light inextensible string. Particle B hangs directly below particle A. A force of P N is applied vertically upwards to particle A, causing the particles to accelerate at 2.5 m s^{-2}.

a. Find the magnitude of force P. (3 marks)

b. Find the tension in the string. (2 marks)

8. E/P A lorry of mass 3000 kg is towing a trailer of mass 5000 kg along a straight horizontal road. The lorry and trailer are connected by a light inextensible tow-bar. The lorry exerts a driving force of 72000 N causing the lorry and trailer to accelerate at 4.5 m s^{-2}. The lorry and trailer experience resistances of $2k$ N and $3k$ N respectively.

a. Find the value of k. (3 marks)

b. Find the tension in the tow-bar. (2 marks)

When the lorry and trailer are moving at 8 m s^{-1}, the tow-bar breaks.

c. Assuming that the driving force on the lorry and the resistances to motion are unchanged, find the distance travelled by the trailer before it comes to rest. (6 marks)

d. State how you have used the modelling assumption that the tow-rope is inextensible. (1 mark)

9. E/P A train consisting of an engine, mass 3200 kg, and a carriage, mass 1500 kg, moves along a straight horizontal track.

The engine is connected to the carriage by an inextensible shunt, which is parallel to the direction of motion of the train. The horizontal resistances to motion of the engine and the carriage have magnitudes 1000 N and 600 N respectively. The engine produces a constant horizontal driving force of magnitude 8650 N that causes the train to accelerate. Find:

a. the acceleration of the train (3 marks)

b. the magnitude of the tension in the shunt. (3 marks)

The train must stop at the next station, so the driver reduces the driving force to zero and applies the brakes. The brakes produce a constant force on the train of magnitude 2500 N, causing the engine and carriage to decelerate.

c. Given that the other resistances to motion are unchanged, find the magnitude of the thrust in the shunt. Give your answer correct to 3 significant figures. (7 marks)

10.6 Pulleys

Practice Book, printed page 74.

1. Particles A and B, of masses m and $3m$, are attached to the ends of a light inextensible string. The string is taut and passes over a small smooth fixed pulley. The system is released from rest.

a.

i. Write an equation of motion for A.

ii. Write an equation of motion for B.

b. Find the magnitude of the acceleration of the particles. Give your answer in terms of g.

c. State how you have used the information that the string is inextensible.

Hint

Draw a diagram showing all the forces acting on each mass and the pulley, and the acceleration.

2. E/P Two particles have masses 4 kg and m kg, where $m < 4$. They are attached to the ends of a light inextensible string. The string passes over a smooth fixed pulley. The particles are held in position with the string taut and the hanging parts of the string vertical.



The particles are then released from rest. The initial acceleration of each particle has magnitude $\frac{3}{13}g$. Find:

a. the tension in the string immediately after the particles are released (3 marks)

b. the value of m. (3 marks)

Hint

You cannot treat a system involving a pulley as a single particle, since the particles are moving in different directions.

3. E/P Two particles A and B, of masses 5 kg and 2 kg respectively, are connected by a light inextensible string, which is taut and passes over a smooth fixed pulley. The particles are released from rest.

a. Giving your answers in terms of g, find:

i. the acceleration of the system

ii. the tension in the string.

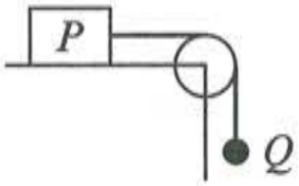
(4 marks)

When A has descended for 2.5 seconds it strikes the ground and immediately comes to rest.

b. Find the speed of A when it hits the ground. (3 marks)

c. Assuming that B does not hit the pulley, find the greatest height that B reaches above its initial position. (3 marks)

4. E/P Two particles P and Q, of masses 6 kg and 5 kg respectively, are connected by a light inextensible string. Particle P lies on a rough horizontal table and the string passes over a small smooth pulley which is fixed at the edge of the table. Particle Q hangs freely.



The friction between P and the table is k N. The system is released from rest and accelerates at a rate of 2 m s^{-2}. Find:

a. the tension in the string (4 marks)

b. the value of k (3 marks)

c. the magnitude of the force exerted on the pulley by the string. (4 marks)

5. E/P Two particles A and B, of masses 2.8 kg and 1.6 kg respectively, are connected by a light inextensible string, which is taut. Particle A lies on a rough horizontal table, 2 m from a small smooth pulley fixed at the edge. The string passes over the pulley and B hangs freely, 0.7 m above ground. A frictional force of magnitude $0.5g$ N opposes the motion of particle A.
The system is released from rest. Find:

a. the acceleration of the system (3 marks)

b. the time taken for B to reach the ground (4 marks)

c. the total distance travelled by A before it first comes to rest. (7 marks)

Problem solving: Set A

Bronze

A car of mass 1200 kg is towing a trailer of mass 1000 kg along a straight horizontal road. The particles are connected by a tow-bar which is modelled as a light rod parallel to their direction of motion. The car is subject to a constant driving force of 4000 N. The resistances to the motion of the car and trailer are constant forces of magnitude 400 N and R N respectively.
The acceleration of the system is 1.5 m s^{-2}.

a. Show that $R = 300$. (3 marks)

b. Find the tension in the tow-bar. (3 marks)

Silver

A train engine of mass 1300 kg is towing a carriage of mass 1100 kg along a straight horizontal track. The carriage is connected to the engine by a shunt, which is parallel to the direction of motion of the train. The train driver applies the brakes, which produces a force on the train of 400 N, and the train decelerates. The resistances to the motion of the engine and the carriage are modelled as constant forces of magnitudes 300 N and 500 N respectively.

a. Find the time taken for the train to decelerate from 12 m s^{-1} to rest. (6 marks)

b. Comment on the assumption that the total resistances to motion are constant. (1 mark)

Gold

A car of mass 1400 kg is towing a caravan of mass 1600 kg along a straight horizontal road. The caravan is connected to the car by a tow-bar which is parallel to the direction of motion of the car and caravan. The tow-bar is modelled as a light inextensible rod.
The driver brakes, producing a force on the car of F N, and the car and trailer decelerate. The resistances to the motion of the car and caravan are modelled as constant forces of magnitude 400 N and 600 N respectively. The magnitude of the thrust in the tow-bar is 200 N.

a. Find the value of F. (6 marks)

The model is refined to account for the mass of the tow-bar. The other values given in the question are unchanged.

b. State how this will affect your answer to part **a**. (1 mark)

Problem solving: Set B

Bronze

Two particles A and B have masses 2 kg and 7 kg respectively. The particles are connected by a light inextensible string which passes over a smooth light fixed pulley. The system is released from rest.

a.

i. Write down an equation of motion for A.

ii. Write down an equation of motion for B.

(3 marks)

b. Find the acceleration. Give your answer in terms of g. (1 mark)

c. State how you have used the information that the pulley is smooth. (1 mark)

Silver

Two particles A and B have masses 1 kg and 5 kg respectively. The particles are connected by a light inextensible string which passes over a smooth light fixed pulley. The particles are initially 0.75 m above the ground. The system is released from rest. Assuming that B hits the ground before A reaches the pulley, find:

a. the speed of B when it hits the ground (6 marks)

b. the time taken for B to hit the ground. (2 marks)

Gold

Two particles A and B have masses 4 kg and 6 kg respectively. The particles are connected by a light inextensible string which passes over a smooth light fixed pulley. The system is released from rest. After descending 2 m, B strikes the ground and is immediately brought to rest. Given that A does not reach the pulley, find the distance travelled by A between the instant when B strikes the ground and the instant when the string next becomes taut. (10 marks)

Selected exam question bank questions

Question 38

Two particles A and B of masses 10 kg and m kg respectively are connected by a light inextensible string. Particle B hangs directly below particle A. A force of 150 N is applied vertically upwards to particle A, causing the particles to accelerate at 3 m s^{-2}.

a. Find the value of m. (3 marks)

b. Find the tension in the string. (2 marks)

Question 41

Two forces, $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$ N and $\begin{pmatrix} p \\ q \end{pmatrix}$ N, act on a particle P.

The resultant of the two forces is **R**. Given that **R** acts in a direction which is parallel to the vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$, show that $4p + 3q + 5 = 0$. (4 marks)

Question 44

A car with mass 1800 kg accelerates from rest to a speed of 12 m s^{-1} in 8 seconds along a horizontal road. The car experiences a constant resistive force of 1400 N. Find:

a. the acceleration of the car (3 marks)

b. the driving force provided by the engine of the car. (2 marks)

c. Comment on the assumption that the total resistances to motion are constant. (1 mark)

Question 47

A particle of mass 0.8 kg lies at rest in a horizontal plane, acted upon by three forces

$$\mathbf{F}_1 = \begin{pmatrix} a \\ b \end{pmatrix} \text{ N}, \quad \mathbf{F}_2 = \begin{pmatrix} b \\ -2a \end{pmatrix} \text{ N and } \mathbf{F}_3 = \begin{pmatrix} -7 \\ 8 \end{pmatrix} \text{ N}$$

a. Find the values of a and b. (3 marks)

The force $\mathbf{F}_3$ is removed. The resultant of the forces $\mathbf{F}_1$ and $\mathbf{F}_2$ is **R**. (2 marks)

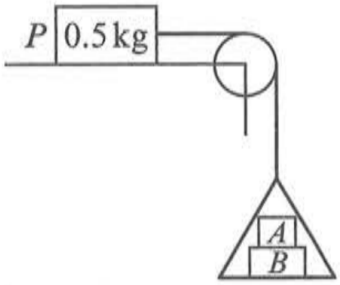
b. Calculate, in newtons, the exact magnitude of **R**.

c. Calculate, to the nearest degree, the angle between the line of action of **R** and the unit vector **i**. (3 marks)

d. Find the distance travelled by the particle in the first 3 seconds after the force $\mathbf{F}_3$ is removed. (3 marks)

Question 53

Block P, of mass 0.5 kg, sits on a rough horizontal table. A light inextensible string passes over a smooth pulley and is attached to block P at one end and a light scale pan at the other, as shown. The scale pan carries two blocks A and B, of mass 0.2 kg and 0.8 kg respectively, with A on top of B. The total resistance to motion of P is modelled as a force of constant magnitude 2 N.



a.

i. Write an equation of motion for P.

ii. Write an equation of motion for the scale pan.

(4 marks)

b. Hence find the:

i. magnitude of the tension in the string

ii. acceleration of P.

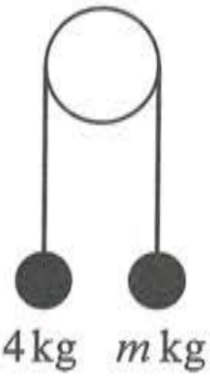
(3 marks)

c. Find the magnitude of the force exerted by block B on block A. (3 marks)

d. State how you have used the fact that the pulley is smooth in your calculations. (1 mark)

Question 57

Two particles A and B have masses 4 kg and m kg respectively, where $m < 4$. The particles are connected by a light inextensible string which passes over a smooth light fixed pulley. The system is held at rest with the string taut. The hanging parts of the string are vertical and A and B are at the same height above a horizontal plane.



The system is released from rest and A descends with acceleration $\frac{3}{5}g$ m s^{-2}.

a. Find the tension in the string. (3 marks)

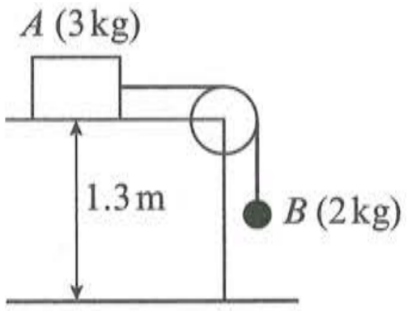
b. Find the value of m. (3 marks)

After descending for 0.6 seconds, particle A reaches the plane and is immediately brought to rest by the impact. The initial distance between B and the pulley is such that B never reaches the pulley.

c. Find the greatest height reached by B above the plane, to 3 significant figures. (7 marks)

Question 59

Particle A of mass 3 kg lies on a smooth horizontal table which is 1.3 m above the floor. A light inextensible string of length 1 m connects A to a particle B of mass 2 kg. The string passes over a smooth pulley which is fixed to the edge of the table and B hangs vertically 0.5 m below the pulley.



The system is released from rest.

a. Find the magnitude of the force exerted on the pulley. (7 marks)

b. Find the speed with which A hits the pulley. (3 marks)

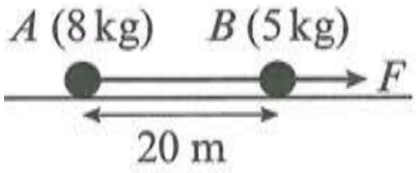
c. State how you have used the fact that the pulley is smooth in your calculations. (1 mark)

When A hits the pulley, the string breaks and B falls to the ground.

d. Find the time between the instant that the string breaks and the instant when B strikes the ground. (4 marks)

Question 60

Two particles A and B of masses 8 kg and 5 kg respectively are initially at rest on a rough horizontal surface. They are joined by a light inextensible string of length 20 m. A constant force F of magnitude 300 N is applied for 2 seconds, during which time particle B travels a distance of 28.8 m.



The total resistances to motion acting on each particle are modelled as being constant and of magnitude k N (so that the total resistance to motion of the whole system has magnitude $2k$ N).

a. Find the value of k. (5 marks)

b. Find the magnitude of the tension in the string. (3 marks)

c. State how you have used the fact that the string is inextensible in your calculations. (1 mark)

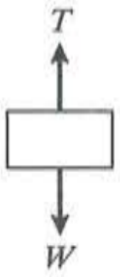
After 2 seconds, the force F is removed and the particles continue to travel in the same straight line.

d. Show that, in the subsequent motion, particle A will collide with particle B. (7 marks)

Answers

10.1 Force diagrams

1.



2. 12 N

3.

a. $P = 15$ N
b. $P = 16$ N, $Q = 24$ N
c. $P = 20$ N, $Q = 45$ N

4.

a. $p = 8$, $q = 5$
b. $p = 15$, $q = 10$
c. $p = 55$, $q = 70$

5.

a.

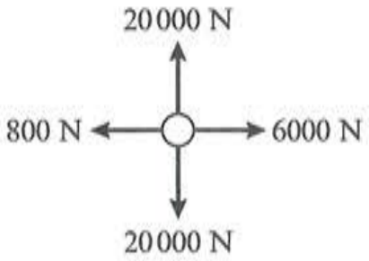
i. $\rightarrow 15$ N
ii. Particle will accelerate to the right

b.

i. $\downarrow 25$ N
ii. Particle will accelerate downwards

6.

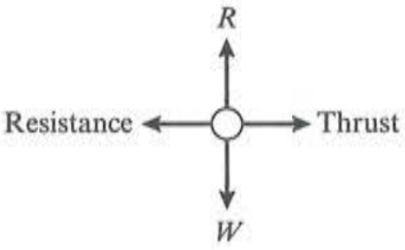
a.



b. 5200 N forwards

7.

a.



b. 3750 N

10.2 Forces as vectors

1.

a. $(\mathbf{i} + \mathbf{j})$ N

b. $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ N

c. $(3\mathbf{i} + 2\mathbf{j})$ N

d. $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ N

2. $a = -5$, $b = -1$

3. $a = 3$, $b = -1$

4. $a = 4$, $b = \frac{8}{3}$

5.

a.

i. $\sqrt{29}$ N

ii. 21.8° (1 d.p.)

b.

i. 5 N

ii. 36.9° (1 d.p.)

c.

i. $\sqrt{41}$ N

ii. 141.3° (1 d.p.)

d.

i. $2\sqrt{10}$ N

ii. 161.6° (1 d.p.)

6.

a.

i. $(4\mathbf{i} - 7\mathbf{j})$ N

ii. $\sqrt{65}$ N

iii. 150.3° (1 d.p.)

b.

i. $(3\mathbf{i} - \mathbf{j})$ N

ii. $\sqrt{10}$ N

iii. 108.4° (1 d.p.)

7.

a. $a = 4$, $b = 3$
b. $\sqrt{122}$ N
c. 5°

8.

a. $a = 4$, $b = 2$
b. $12\sqrt{2}$ N
c. 135°

9. 5

10.3 Forces and acceleration

1. 0.64 m s^{-2}

2. 24 N

3. 88.2 N

4.

a. 0.3 m s^{-2}
b. 10 seconds

5.

a. 71.2 N
b. 29.4 N
c. 51 N

6.

a. 3.125 kg
b. 1.72 kg (3 s.f.)
c. 2.85 kg (3 s.f.)

7.

a. 5.2
b. 7.93 (3 s.f.)
c. 2.7

8. 20 N

9.

a. 588 N
b. 618 N

10.

a. 1.2 m s^{-2}
b. 9000 N

11.

a. 120
b. 2.25 m

10.4 Motion in 2 dimensions

1.

a. $\left(\frac{3}{2}\mathbf{i} + \frac{1}{4}\mathbf{j}\right)$ m s^{-2}

b. $\frac{\sqrt{37}}{4}$ m s^{-2} at 081° (to the nearest degree)

2. 0.25

3.

a. $(50\mathbf{i} - 30\mathbf{j})$ N

b. $10\sqrt{34}$ N at 121°

4.

a. $(2\mathbf{i} + 22\mathbf{j})$ m s^{-2}

b. $\left(\frac{11}{7}\mathbf{i} - \frac{2}{7}\mathbf{j}\right)$ m s^{-2}

c. $\left(-\frac{5}{4}\mathbf{i} - \frac{25}{12}\mathbf{j}\right)$ m s^{-2}

d. $\left(\frac{6}{5}\mathbf{i} - 4\mathbf{j}\right)$ m s^{-2}

5.

a. 1 m s^{-2} at −36.9° from **i**

b. 2 seconds

c. 4.47 m s^{-1} (2 s.f.)

6. $-2(p+1) = q - 5 \Rightarrow -2p - 2 = q - 5 \Rightarrow 2p + q - 3 = 0$

7. $p = 8$, $q = -4$

8.

a. $9\sqrt{3}$

b. 18 N

c. 6 m s^{-2}

d. 5 seconds

9.

a. $p = 5$, $q = -12$

b. $\frac{5\sqrt{50}}{13}$ kg

10. 4

11.

a. 63.4° (3 s.f.)

b. $2(q-2) = (6+p) \Rightarrow 2q - 4 = 6 + p \Rightarrow p - 2q + 10 = 0$

c. 2.5

10.5 Connected particles

1.

a. 6.4 N

b. 4 N

2.

a. 12

b. 20 N

3.

a. 2

b. 30 N

4.

a. 21500 N

b.

i. 1290 N upwards

ii. 3440 N downwards

5.

a. Lorry 3200 kg, trailer 8000 kg

b. 44000 N

6.

a. 0.2

b. 90 N

7.

a. 332 N (3 s.f.)

b. 148 N (3 s.f.)

8.

a. 7200

b. 44100 N

c. 7.41 m

d. Treating the tow-bar as inextensible is to assume that the acceleration of the lorry and the trailer are the same.

9.

a. 1.5 m s^{-2}

b. 2850 N

c. 709 N (3 s.f.)

10.6 Pulleys

1.

a.

i. $T - mg = ma$

ii. $3mg - T = 3ma$

b. $0.5g$ m s^{-2}

c. The magnitude of acceleration of the two particles is the same.

2.

a. $\frac{40}{13}g$ N

b. 2.5

3.

a.

i. $\frac{3}{7}g$ m s^{-2}

ii. $\frac{20}{7}g$ N

b. 10.5 m s^{-1}

c. 18.75 m

4.

a. 39 N

b. 27

c. 55.2 N (3 s.f.)

5.

a. $0.25g$ m s^{-2}

b. 0.756 s (3 s.f.)

c. 1.68 m (3 s.f.)

Problem solving: Set A

Bronze

a. $4000 - 400 - R = 2200 \times 1.5 \Rightarrow 3600 - R = 3300$
$\Rightarrow R = 3600 - 3300 = 300$ N

b. 1800 N

Silver

a. 24 seconds

b. Unlikely, as resistance will decrease as speed decreases.

Gold

a. 500 N

b. F will increase.

Problem solving: Set B

Bronze

a.

i. $T - 2g = 2a$

ii. $7g - T = 7a$

b. $a = \frac{5}{9}g$ m s^{-2}

c. This means that you can assume that the tension in the string will be the same on both sides of the pulley.

Silver

a. 3.13 m s^{-1} (3 s.f.)

b. 0.479 seconds (3 s.f.)

Gold 0.8 m

Selected exam question bank answers

Question 38

a. 1.7 (2 s.f.)
b. 22 N

Question 41 $-4(p-1) = 3(q+3) \Rightarrow -4p + 4 = 3q + 9 \Rightarrow 4p + 3q + 5 = 0$

Question 44

a. 1.5 m s^{-2}
b. 4100 N
c. Unlikely, as resistance to motion will increase as velocity increases.

Question 47

a. $a = 5$, $b = 2$
b. $\sqrt{113}$ N
c. 49°
d. 59.8 m

Question 53

a.
  i. $T - 2 = 0.5a$
  ii. $g - T = a$
b.
  i. 4.6 N
  ii. 5.2 m s^{-2}
c. 0.92 N
d. The tension in the string is the same on both sides of the pulley.

Question 57

a. $\frac{8}{5}g$ N
b. 1
c. 2.75 m (3 s.f.)

Question 59

a. 16.6 N
b. 1.98 m s^{-1}
c. Tension is equal throughout the system.
d. 0.250 s (3 s.f.)

Question 60

a. $k = 56.4$

b. 171.6 N

c. Each particle (and the whole system) have the same acceleration.

d. Speed of system after 2 seconds is 28.8 m s^{-1}.
For A: $-56.4 = 8a \Rightarrow a = -7.05$ m s^{-2}
For B: $-56.4 = 5a \Rightarrow a = -11.28$ m s^{-2}
Using $v^2 = u^2 + 2as$, A would stop in 58.8 m (3 s.f.) and B would stop in 36.8 m (3 s.f.).
Since $36.8 + 20 < 58.8$, the particles will collide.